# Chi-Square Distribution

# Recap

Theoretical probability distributions are regulated by **parameters**.

- $X \sim N(\mu, \sigma)$
- $X \sim B(N, \theta)$

These parameters are sometimes independent (normal) and sometimes associated with one another (binomial)

# Our Road Map

- What is a model?
- Is our model good?
- How would we know if our model is good?
- Describe your data
- Calculate some measure of center and spread of distributions
- Start to ask hypothesis-based questions
- Calculate some test statistic
- How likely is it that we would obtain our test statistic or more extreme, given the null hypothesis is true? Does it cross some threshold?
- Acknowledge that this process sucks
- Is this the only way of knowing if our model is good?

# "Goodness of Fit"

**Does our model fit the data?**

If it does...?

If it doesn't...?

Enter the $\chi^2$ distribution

# What have we been doing this semester?

- Take a normally distributed variable
- Get deviation scores
- (Standardize them -- z-scores)
- Square the deviation scores
- Sum them up

# Chi-Square Distribution

$\chi^2 = \sum_{i=1}^{k} Z_i^2$

- $Z_i$ = standard normal variable
- $k$ = number of independent z-scores being squared (**degrees of freedom** of the $\chi^2$ distribution)

In English:

> $\chi^2$ is created by squaring standard normal variables and adding them together

# Chi-Square Distribution

$$\chi^2 = \sum_{i=1}^{k} Z_i^2$$

- $Z_i$ = standard normal variable
- $k$ = number of independent z-scores being squared (**degrees of freedom** of the $\chi^2$ distribution)

1. Take a bunch of random z-scores
2. Square each one so they're all positive
3. Add them up.

Do that over and over, and the totals you get follow the $\chi^2$ distribution.

# Chi-Square Distribution

The **chi-square** ( $\chi^2$ ) distribution is unique in that we really only have one parameter, $k$, which are your **degrees of freedom** of the $\chi^2$ distribution

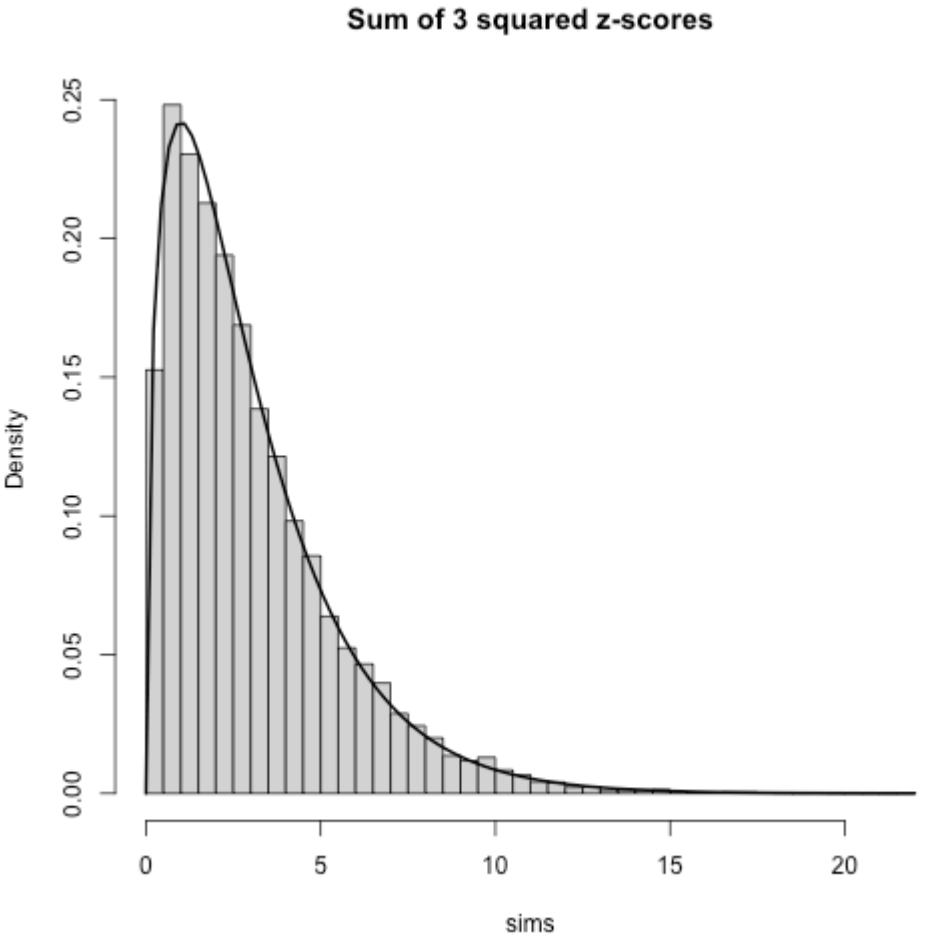
- mean = $k$
- variance = $2k$

It is entirely defined by degrees of freedom! It is the *number of pieces of information that you have*

# What kind of crazy function would lead to a distribution like this?!?!!

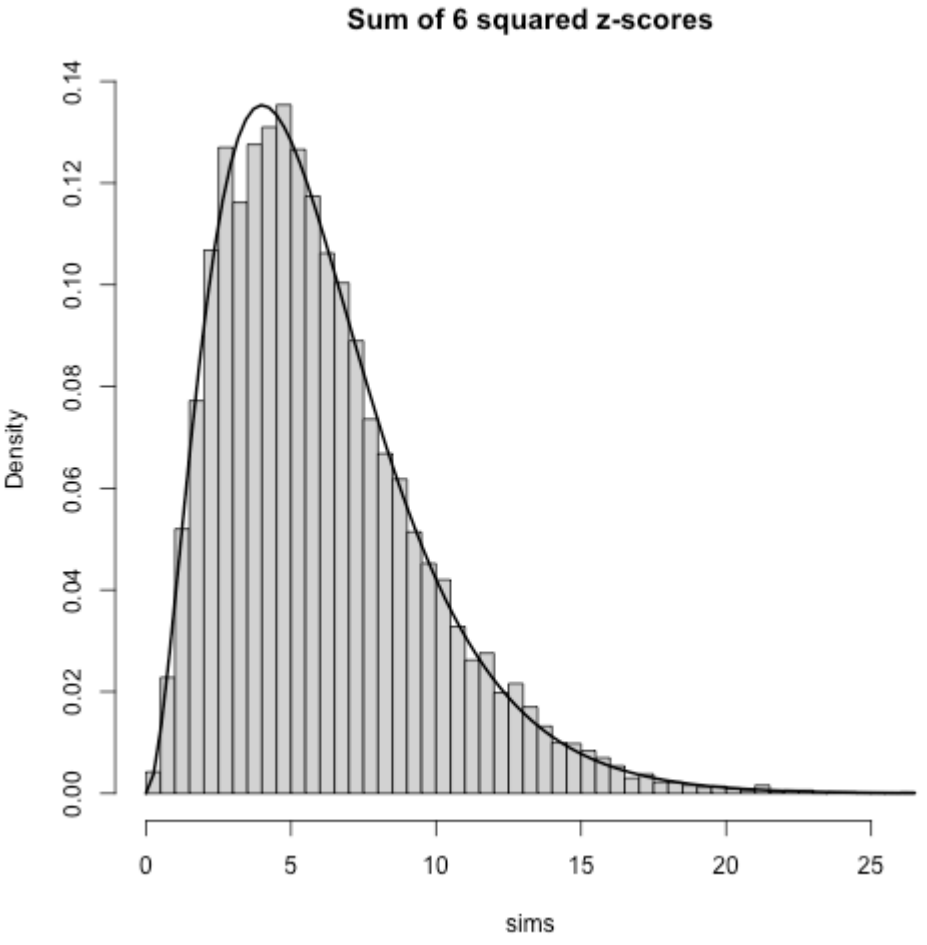
$$\frac{1}{2^{\frac{k}{2}}\Gamma\left(\frac{k}{2}\right)}\, x^{\frac{k}{2}-1} e^{-\frac{x}{2}}$$





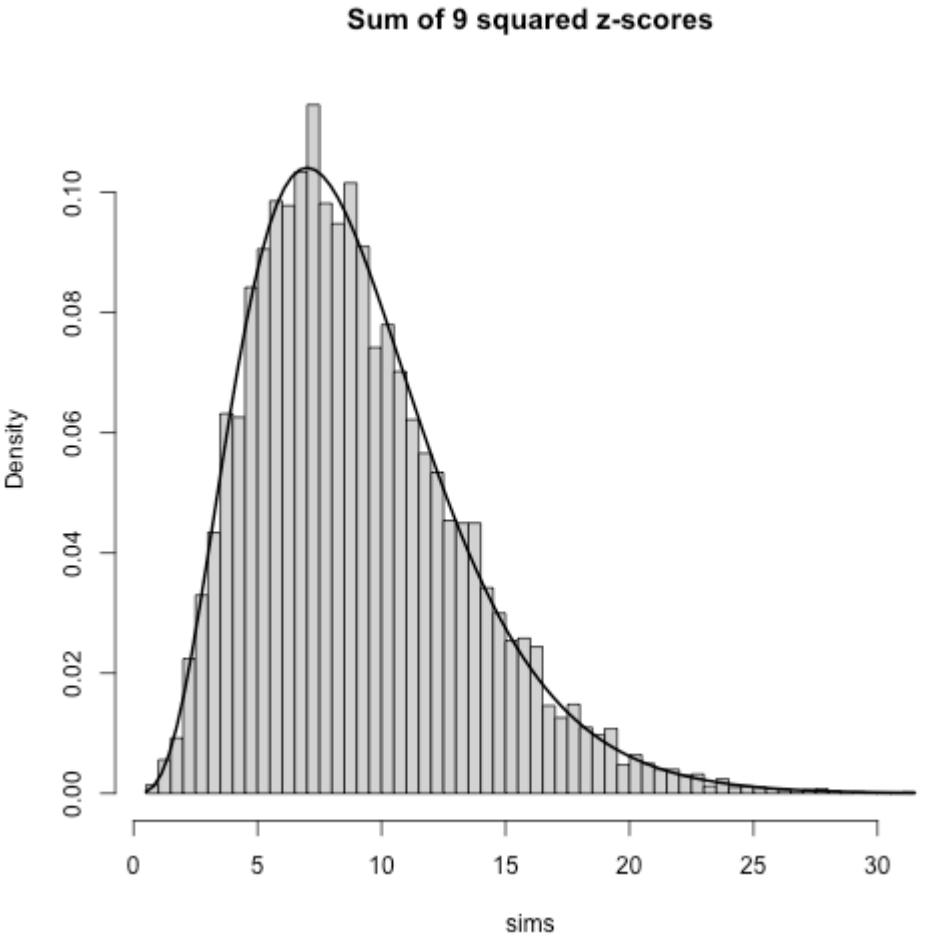

```
## [1] 3.021785
```

```
## [1] 5.928419
```



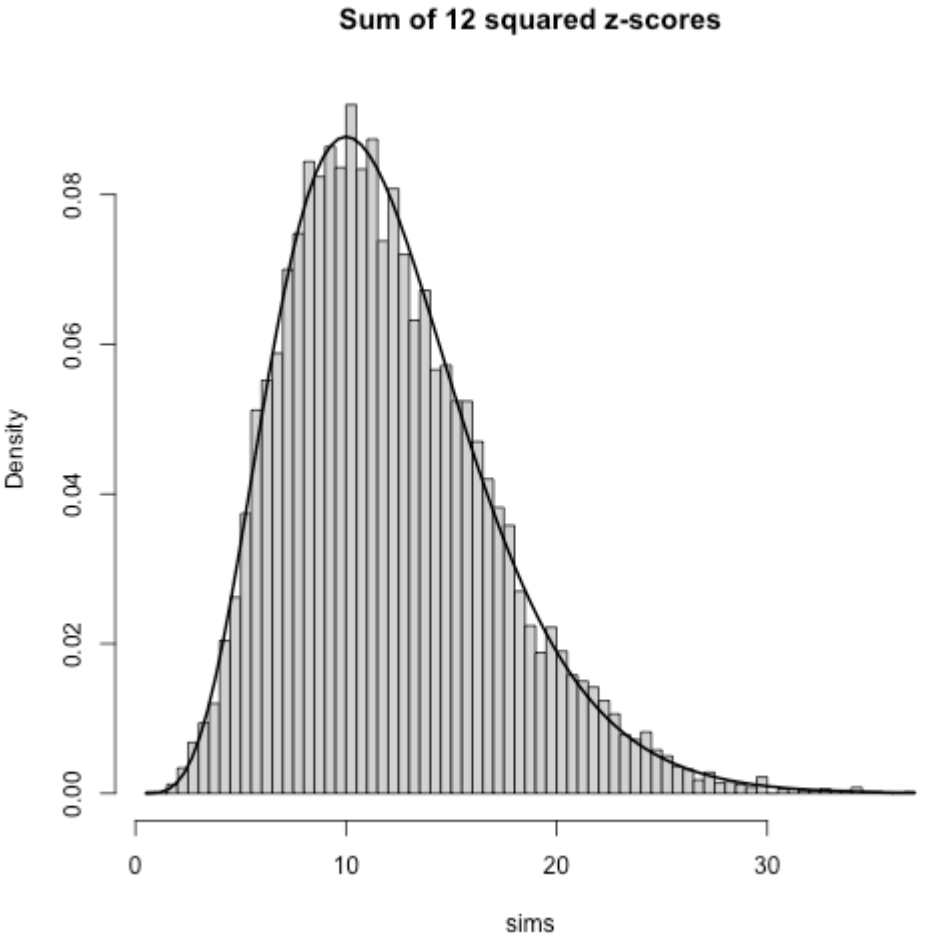

```
## [1] 6.047744
```

```
## [1] 12.10487
```



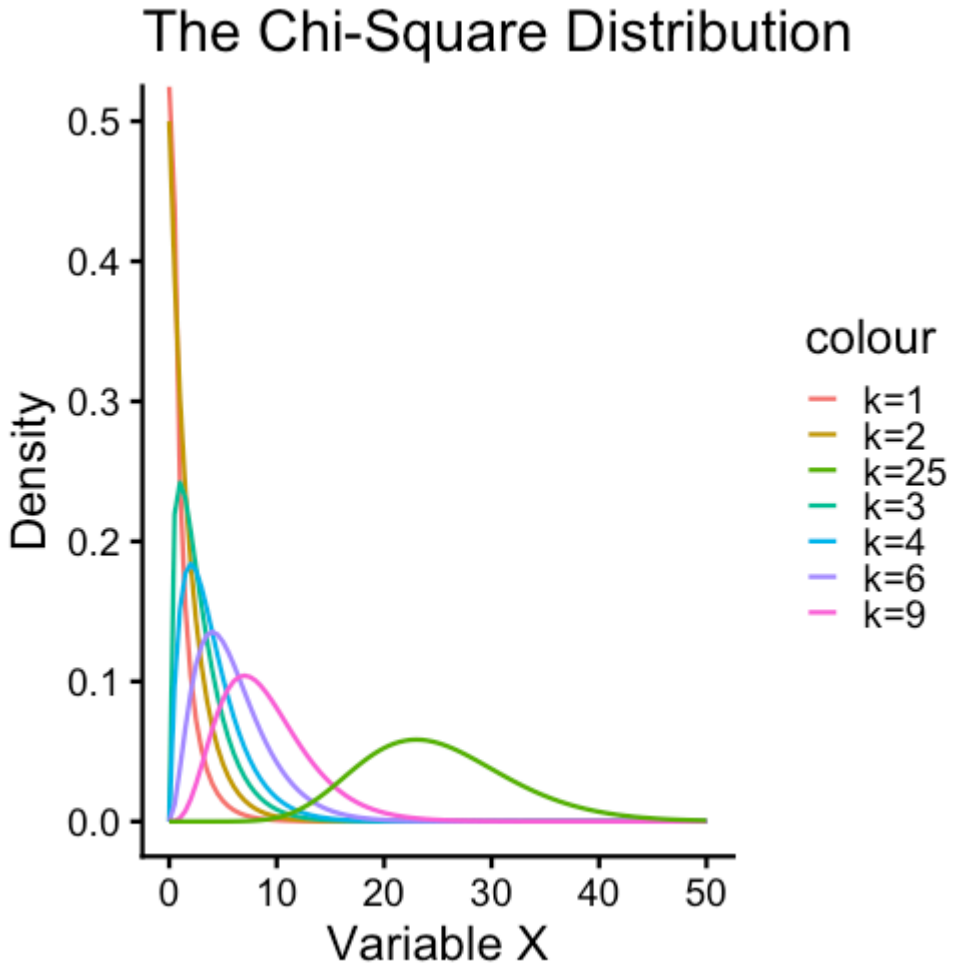

```
## [1] 9.05063
```

```
## [1] 18.04789
```




```
## [1] 12.06
```

```
## [1] 24.39804
```


The Chi-Square Distribution
Density
0.5
0.4
0.3
0.2
0.1
0.0
0
10
20
30
40
50
Variable X
colour
k=1
k=2
k=25
k=3
k=4
k=6
k=9

# Goodness of Fit

Let's go back to our idea that we create a model of how the world works. **It is not perfect**.

If our model fits our observed data, then we can say *"hey, this might be useful for understanding how the world works"*.

If it doesn't fit the data well, then we say *"this probably isn't that helpful, we should revise or re-think our theories of how the world works"*

"All models are wrong, but some are useful" - Box

Suppose psychological theory predicts that responses should be equally distributed across four categories:

- Category A: 25%
- Category B: 25%
- Category C: 25%
- Category D: 25%

We collect data on this variable and observe the following proportions:

- Category A: 30%
- Category B: 24%
- Category C: 23%
- Category D: 23%

**Question: Is this distribution significantly different from what our theory predicts?**

**What information would you need to answer this?**

```r
# Expected proportions under the null hypothesis
expected_prop <- c(A = 0.25, B = 0.25,
                   C = 0.25, D = 0.25)

# Observed proportions
observed_prop <- c(A = 0.30, B = 0.24,
                   C = 0.23, D = 0.23)

# Four sample sizes
sample_sizes <- c(100, 400, 4000, 40000)

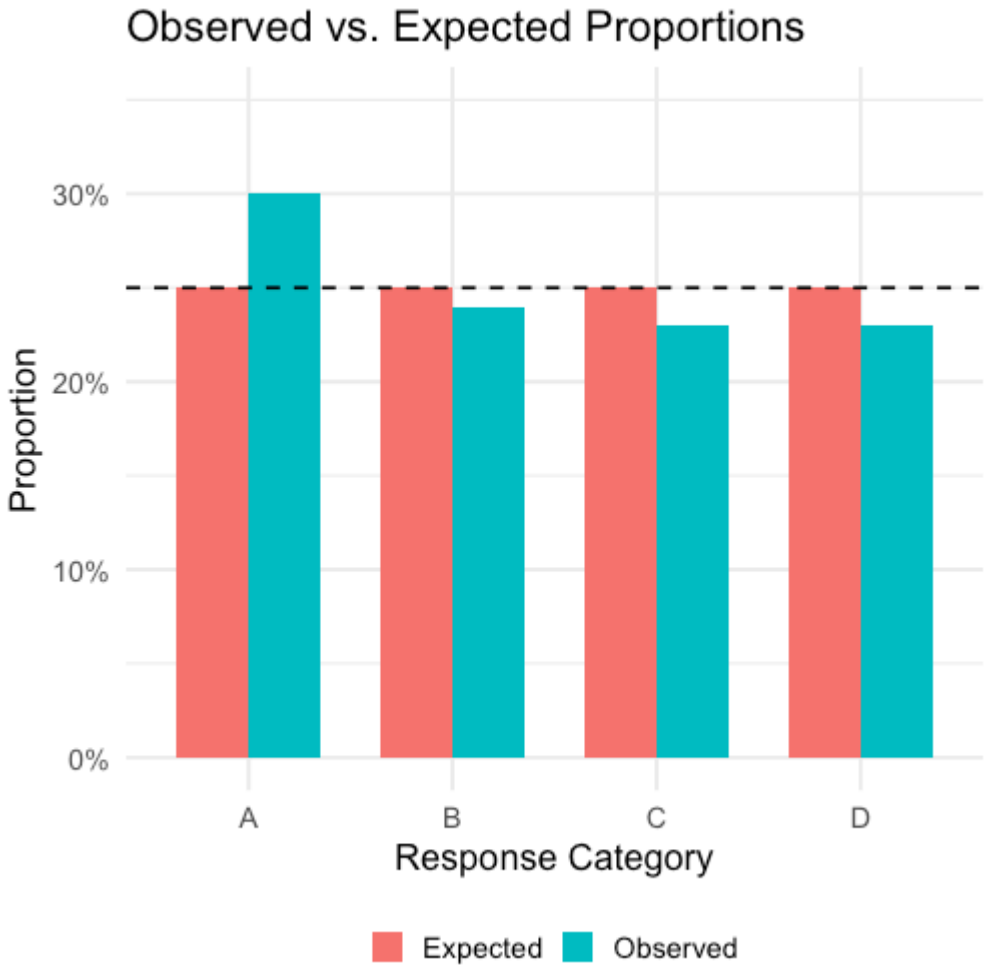
# Construct observed counts at each sample size
data_by_n <- lapply(sample_sizes, function(n) {
  observed_prop * n
})

names(data_by_n) <- paste0("N = ", sample_sizes)

# Display observed counts for each N
do.call(rbind, data_by_n)
```

```
##                A    B    C    D
## N = 100       30   24   23   23
## N = 400      120   96   92   92
## N = 4000    1200  960  920  920
## N = 40000 12000 9600 9200 9200
```

The observed proportions are always 30%, 24%, 23%, and 23%.




**Key Question**: What does this figure look like at N = 100, 400, 4000, 40000?

# A test!

$H_0$ = The four categories have equal probabilities of occurring. $p_a = p_b = p_c = p_d = .25$

$H_A$ = The four categories do not have equal probabilities of occurring. At least 1 $p_i \neq .25$

$$\chi^2 = \Sigma \frac{(O_i - E_i)^2}{E_i}$$

$$\chi^2 = \Sigma \frac{(\text{observed frequency} - \text{expected frequency})^2}{\text{expected frequency}}$$

- $O_i$ = observed count
- $E_i$ = expected count under the null hypothesis. $N/4$

# Why is this a Chi-square?

For a count, the variance is roughly the expected count. So $\frac{(O_i - E_i)}{\sqrt{E_i}}$ acts like a z-score for each category (it's approximate though). Square those and add them up, and you get a sum of squared z-scores, which is a $\chi^2$.

You lose one df because the counts must add up to N. Once you know three of the counts, the fourth is fixed. That's where j − 1 comes from

Note: this breaks a little when expected counts are too small (less than 5 per category). Come find me if you run into this situation

# Chi-Square Goodness of Fit: N = 100

Our null hypothesis is that responses are equally distributed across four categories.

| Category | Observed (O) | Expected (E) | $(O - E)^2 / E$ |
|---|---|---|---|
| A | 30 | 25 | |
| B | 24 | 25 | |
| C | 23 | 25 | |
| D | 23 | 25 | |
| **Total** | **100** | **100** | **?** |

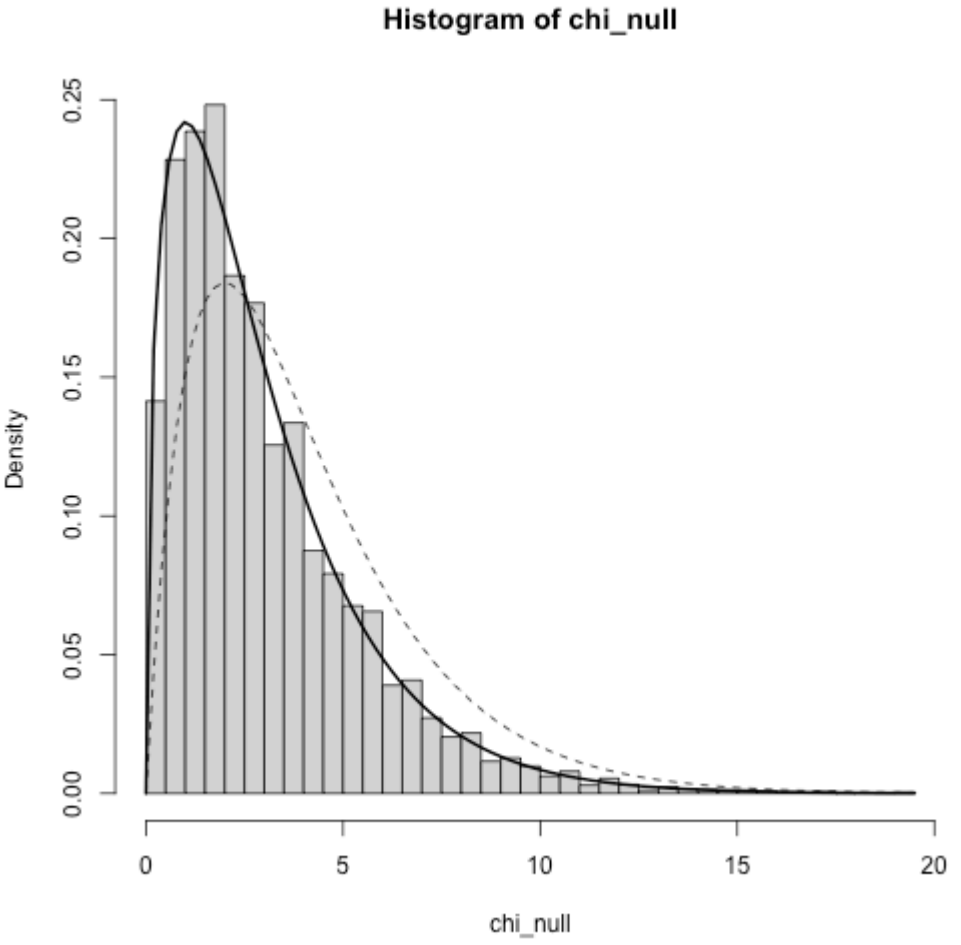
# Chi-Square Goodness of Fit: N = 100

Calculate the contribution of each category to the chi-square statistic.

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

1. What is the total chi-square statistic?
2. What are the degrees of freedom?
3. What is the p-value?
4. At $\alpha = .05$, do we reject the null hypothesis?

- 1.36
- j - 1 = 3, j = number of categories; 4 - 1 = 3
- `pchisq(1.36, df = 3, lower.tail = FALSE)` = .714
- No




10,000 simulations under the null hypothesis (.25 per category). It looks much more like the solid line (df = 3) than the dashed line (df = 4)

# Chi-Square Goodness of Fit: N = 400

| Category | Observed (O) | Expected (E) | $(O - E)^2 / E$ |
|---|---|---|---|
| A | 120 | 100 | |
| B | 96 | 100 | |
| C | 92 | 100 | |
| D | 92 | 100 | |
| **Total** | **400** | **400** | **?** |

Calculate the chi-square statistic, degrees of freedom, and p-value.

**Before calculating:** Do you expect the p-value to be larger or smaller than it was for $N = 100$? Why?

# Chi-Square Goodness of Fit: N = 400

Calculate the contribution of each category to the chi-square statistic.

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

1. What is the total chi-square statistic?
2. What are the degrees of freedom?
3. What is the p-value?
4. At $\alpha = .05$, do we reject the null hypothesis?

- 5.44
- j - 1 = 3
- `pchisq(5.44, df = 3, lower.tail = FALSE)` = .142
- No

# Chi-Square Goodness of Fit: N = 4,000

| Category | Observed (O) | Expected (E) | $(O - E)^2 / E$ |
|---|---|---|---|
| A | 1,200 | 1,000 | |
| B | 960 | 1,000 | |
| C | 920 | 1,000 | |
| D | 920 | 1,000 | |
| **Total** | **4,000** | **4,000** | **?** |

Calculate the chi-square statistic and p-value.

**Is the discrepancy between the observed and expected proportions any different from the discrepancy at N = 100? What about the statistical conclusion?**

# Chi-Square Goodness of Fit: N = 4,000

Calculate the contribution of each category to the chi-square statistic.

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

1. What is the total chi-square statistic?
2. What are the degrees of freedom?
3. What is the p-value?
4. At $\alpha = .05$, do we reject the null hypothesis?

- 54.4
- j - 1 = 3
- `pchisq(54.4, df = 3, lower.tail = FALSE)` = 9.22x10^-12
- **YES**

# Chi-Square Goodness of Fit: N = 40,000

| Category | Observed (O) | Expected (E) | $(O - E)^2 / E$ |
|---|---|---|---|
| A | 12,000 | 10,000 | |
| B | 9,600 | 10,000 | |
| C | 9,200 | 10,000 | |
| D | 9,200 | 10,000 | |
| **Total** | **40,000** | **40,000** | **?** |

Calculate the chi-square statistic and p-value.

**Final question:** Has the distribution become more substantively different from the null hypothesis, or have we simply become more certain that it is not exactly equal to the null distribution?

# Chi-Square Goodness of Fit: N = 40,000

Calculate the contribution of each category to the chi-square statistic.

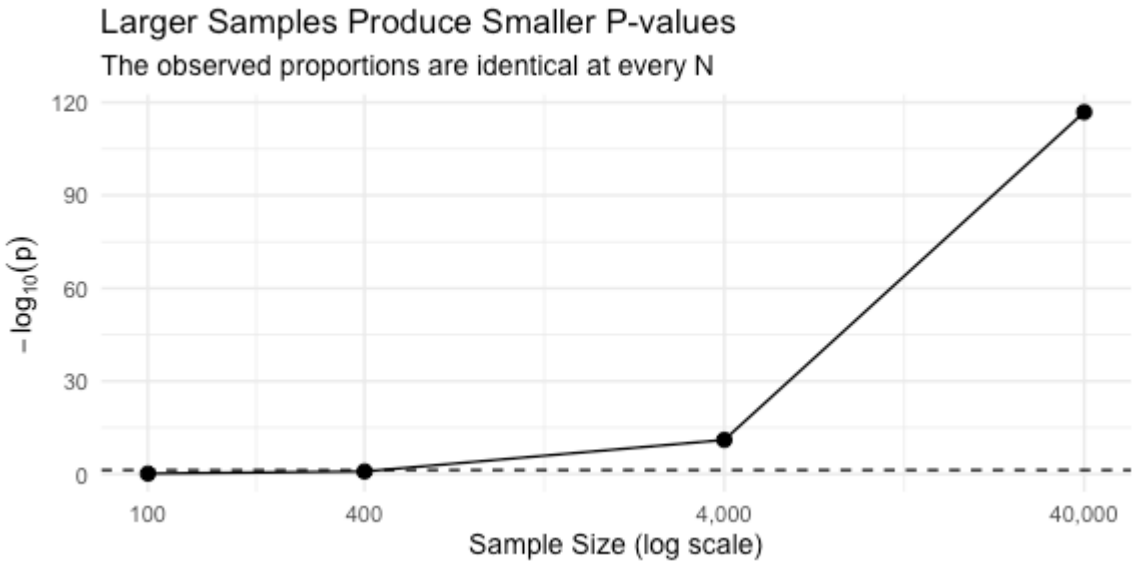
$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

1. What is the total chi-square statistic?
2. What are the degrees of freedom?
3. What is the p-value?
4. At $\alpha = .05$, do we reject the null hypothesis?

- 544.0
- j - 1 = 3
- `pchisq(544.0, df = 3, lower.tail = FALSE)` = 1.39x10^-117
- **YES**

What changed across the 4 tests?

# Visualizing the P-values




The dashed line corresponds to p = .05.

Why does the p-value decrease even though the observed proportions remain unchanged?

Under the null hypothesis, the expected counts are $E_i = Np_i$. With the observed proportions held constant, $O_i = N\hat{p}_i$

- $p_i$ is the expected proportion under the null hypothesis (.25).
- $\hat{p}_i$ is the observed proportion.

$$\chi^2 = \sum_{i=1}^{j} \frac{(O_i - E_i)^2}{E_i}$$

Substituting the counts:

$$\chi^2 = N \sum_{i=1}^{j} \frac{(\hat{p}_i - p_i)^2}{p_i}$$

The summation term is constant because the observed and expected proportions do not change.

Therefore:

$$\chi^2 \propto N$$

In our example:

$$\chi^2 = 0.0136N$$

As N increases, the test statistic increases while the degrees of freedom remain fixed. The p-value therefore decreases.

# The Discrepancy Has Not Changed

```
##   Category Expected Observed Difference
## 1        A     0.25     0.30       0.05
## 2        B     0.25     0.24      -0.01
## 3        C     0.25     0.23      -0.02
## 4        D     0.25     0.23      -0.02
```

The differences from the expected proportions are always:

- A: +5 percentage points
- B: −1 percentage point
- C: −2 percentage points
- D: −2 percentage points

The same proportional discrepancy produces different p-values because the **precision of our estimates changes with N**.

# Is This Difference Important?

1. What was N when we first rejected the null?
2. As N got bigger, did the observed distribution (or proportions) become *more* different than the hypothesized distribution?
3. What additional information would you want when reading a paper reporting this result?
4. If the null model is a useful approximation, does detecting a tiny departure necessarily make the model useless?

If the null hypothesis is even a little bit wrong, a large enough N will find it...



# What Does a Small p-value Tell Us?

A small p-value tells us that the observed data (or more extreme) would be unusual under the very *specific* null model

It does **not** tell us:

- How large the discrepancy is
- Whether the discrepancy is theoretically important
- Whether the null model is a useful approximation
- Whether the effect is practically (or clinically) meaningful.

**Statistical significance $\neq$ practical significance!**

# Interim Summary

The test is not broken. It is correctly detecting departures from the **exact** null hypothesis.

The problem arises when we treat rejection of an exact null hypothesis as proof that a *meaningful* effect exists.

2 separate but related questions to ask:

1. Is the observed data inconsistent with the null model? [Collect $200]
2. Is the observed departure large enough to matter for our scientific question?

# What can we do? Effect sizes!

We've actually already done this (almost)!

Cohen's $w^2 = \sum \frac{(\hat{p}_i - p_i)^2}{p_i}$

Cohen's w = $\sqrt{\sum \frac{(\hat{p}_i - p_i)^2}{p_i}}$

What is Cohen's w when:

- N = 100
- N = 400
- N = 4,000
- N = 40,000

**THEY ARE THE SAME!**

$w = \sqrt{.0136} \approx .12$; small

# Effect Sizes

Here, $\chi^2 = Nw^2$

**Nearly all test statistics are roughly effect size x (some function of) sample size**

# Are Two Variables Related?

Same idea, new question: did **Condition** affect whether people chose **Category A**?

| | Chose A | Didn't | Total |
|---|---|---|---|
| Condition 1 | 30 | 70 | 100 |
| Condition 2 | 20 | 80 | 100 |
| **Total** | **50** | **150** | **200** |

$H_0$: Condition and choice are unrelated

If that's true, both conditions should choose A at the *overall* rate: 50/200 = 25%

$$E = \frac{\text{row total} \times \text{column total}}{N} = \frac{100 \times 50}{200} = 25$$

| | Chose A | Didn't | Total |
|---|---|---|---|
| Condition 1 | 30 | 70 | 100 |
| Condition 2 | 20 | 80 | 100 |
| **Total** | **50** | **150** | **200** |

$$E = \frac{\text{row total} \times \text{column total}}{N} = \frac{100 \times 50}{200} = 25$$

So we *expected* 25 A's per condition, but we observed 30 and 20. Same question as before: **how far is the data from the model?**

# Chi-Square Test of Independence

Same formula, now summed over all 4 cells:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

df = (rows − 1)(columns − 1) = (2 − 1)(2 − 1) = **1**

```
tab <- matrix(c(30, 70,
                20, 80), nrow = 2, byrow = TRUE,
              dimnames = list(Condition = c("1", "2"),
                              ChoseA = c("Yes", "No")))

chisq.test(tab, correct = FALSE)
```

```
##
##     Pearson's Chi-squared test
##
## data:  tab
## X-squared = 2.6667, df = 1, p-value = 0.1025
```

# Chi-Square Test of Independence

Same formula, now summed over all 4 cells:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$\chi^2(1) = 2.67, p = .10$. Not significant.

R applies a "continuity correction" to 2×2 tables by default. It's overly conservative, so we turn it off with `correct = FALSE`.

# Effect Size: The Odds Ratio

**Odds** = (# who did it) / (# who didn't)

- Condition 1: 30/70 = 0.43
- Condition 2: 20/80 = 0.25

**Odds ratio** = 0.43 / 0.25 = **1.71**

> The odds of choosing A were 1.71 times higher in Condition 1.

- OR = 1 → no association
- OR > 1 → higher odds
- OR < 1 → lower odds

## Watch what happens when we multiply every cell by 10:

```
chisq.test(tab * 10, correct = FALSE)
```

```
##
##     Pearson's Chi-squared test
##
## data:  tab * 10
## X-squared = 26.667, df = 1, p-value = 2.418e-07
```

$\chi^2(1) = 26.67, p < .001$, but the OR is **still 1.71**

**Statistical significance $\neq$ practical significance!**

**Spoiler:** logistic regression (spring!) is basically a machine for estimating odds ratios.

# What do we want to know?

Are variables related?

- Chi-square test of association / independence (contingency tables)

Does the data match what a model predicts?

- Goodness-of-fit test

Does a more complex model improve fit?

- Likelihood ratio test

Does the model adequately explain the data?

- Deviance statistic

# What do we want to know?

All of these ask the same underlying question:

> How far are the observed data from what the model predicts?

They are cousins of this formula:

$$\chi^2 = \Sigma \frac{(\text{observed frequency} - \text{expected frequency})^2}{\text{expected frequency}}$$

# Next Time

- t-tests galore!